

Foundation (Jalapeno)

Q1. Factor the expression $x^2 - 4$

- (A) $(x - 2)(x + 2)$
- (B) $x(x - 4)$
- (C) $x(x + 4)$
- (D) $x^2 - 2x + 4$
- (E) $x^2 + 2x - 4$

Q2. Factor the expression $9y^2 - 25$

- (A) $3y(3y - 5)$
- (B) $3y(3y + 5)$
- (C) $(3y - 5)(3y + 5)$
- (D) $9y(y - 5)$
- (E) $9y(y + 5)$

Q3. Which expression is equivalent to $(a - b)(a + b)$?

- (A) $a^2 - b^2$
- (B) $a^2 + b^2$
- (C) $a^2 - 2ab + b^2$
- (D) $a^2 + 2ab + b^2$
- (E) $a^2 - 2a + b^2$

Q4. Factor the expression $64 - p^2$

- (A) $8(8 - p)$
- (B) $8(8 + p)$
- (C) $(8 - p)(8 + p)$
- (D) $64 - 2p + p^2$
- (E) $64 + 2p + p^2$

Q5. Factor the expression $x^2 - 9y^2$

- (A) $(x - 3y)(x + 3y)$
- (B) $(x - 3y)(x - 3y)$
- (C) $(x + 3y)(x + 3y)$
- (D) $x(x - 9y)$
- (E) $x(x + 9y)$

Q6. Which of the following can be factored using the difference of squares formula?

- (A) $x^2 + 9$
- (B) $x^2 - 9$
- (C) $x^3 - 8$
- (D) $x^2 - x - 6$
- (E) $x^2 + x + 1$

Q7. Factor the expression $16p^2 - 49q^2$

- (A) $(4p - 7q)(4p + 7q)$
- (B) $(4p - 7q)(4p - 7q)$
- (C) $(4p + 7q)(4p + 7q)$
- (D) $16p(p - 7q)$
- (E) $16p(p + 7q)$

Q8. Factor the expression $25x^2 - 36y^2$

- (A) $(5x - 6y)(5x + 6y)$
- (B) $(5x - 6y)(5x - 6y)$
- (C) $(5x + 6y)(5x + 6y)$
- (D) $25x(x - 6y)$
- (E) $25x(x + 6y)$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^2 - 64y^2$ and explain your reasoning

Q10. Factor the expression $49m^2 - 16n^2$ and provide a real-world example using this expression

Chef's Tips

- Read all questions carefully before starting
- Use the difference of squares formula only when applicable
- Clearly show your work and explain reasoning in free response questions